

DINETH HASARANGA

FULL-STACK & AI ML SOFTWARE ENGINEER

071-4713270 | hasarangadineth@icloud.com | <https://github.com/dineth-cs> | <https://www.linkedin.com/in/dineth-hasaranga>

SUMMARY

A passionate Computer Science undergraduate specializing in Full-Stack Development, UI/UX Design, and Multimodal Artificial Intelligence. Proven expertise in engineering secure, scalable systems and crafting engaging, user-centric cross-platform mobile experiences. Highly driven to solve complex real-world challenges by combining intuitive interfaces and robust functional system architectures with advanced AI technologies, including CNNs, Transformers, and NLP.

PROJECT EXPERIENCE

MindAura – Multimodal AI Personal Assistant

- Developed a prototype mobile application utilizing React Native to function as a comprehensive lifestyle and health companion.
- Engineered a multimodal AI system integrating CNNs for facial emotion recognition, DNNs for vocal analysis, and Transformer models for text sentiment processing.
- Designed a secure functional system architecture featuring a strict process-and-discard mechanism for biometric inputs, fortified by AES-256 encryption.
- Achieved classification accuracies exceeding 85% on standard validation datasets, including FER-2013 and RAVDESS.

SMS Shield – Real-Time Threat Detection Platform

- Developed a machine learning pipeline using Python, Scikit-learn, and Pandas to identify and block phishing attempts and scam SMS messages in real-time.
- Achieved a 96.2% accuracy rate in threat classification by utilizing a TfidfVectorizer and a Multinomial Naive Bayes classifier.
- Designed and deployed a highly responsive, low-latency frontend interface using Streamlit, featuring customized CSS for a modern glassmorphism UI.

ChurnGuard – Enterprise AI Platform

- Developed an end-to-end Machine Learning solution to predict and mitigate customer churn in the telecommunications industry.
- Engineered a Predictive AI Engine utilizing a Random Forest Classifier trained on 7,000+ customer profiles to achieve high-accuracy predictions.
- Designed and deployed a sleek, "Glassmorphism" inspired enterprise dashboard using Streamlit and Plotly, emphasizing intuitive UI/UX design.
- Implemented a scalable Batch Prediction Module for processing large CSV datasets and a Financial ROI Calculator to translate AI insights into measurable business impact.

NexaMart – Modern Real-Time E-commerce Platform

- Developed a full-featured, real-time e-commerce web application utilizing the MERN Stack (MongoDB, Express, React, Node.js).
- Engineered an Advanced Admin Dashboard with a dynamic CMS, allowing seamless management of products, orders, and custom promotional banners.
- Integrated Socket.io to facilitate real-time inventory updates and instant admin notifications.
- Designed a responsive, premium user interface focusing on modern UI/UX principles, utilizing Tailwind CSS, glassmorphism effects, and interactive product cards.
- Implemented comprehensive backend functionalities including secure user authentication, multi-image uploads with automatic compression, and a complete shopping cart system.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, Dart, C#, MATLAB, HTML/CSS
- Frontend & Mobile: React.js, React Native, Streamlit, Tailwind CSS, Flutter
- Backend & APIs: Node.js, Express.js, FastAPI
- Databases & Tools: MongoDB, Git, GitHub, Socket.io, Figma, Firebase, SQL Server, Cloudinary
- AI & Machine Learning: Scikit-learn, Pandas, OpenCV, Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), Transformers, NLP, Plotly, Random Forest

EDUCATION

BSc (Hons) in Computer Science NSBM Green University | (2023 – 2026)

- Sponsored by / Affiliated with Plymouth University (UK).
- Currently completing major software development projects and academic research focusing on Multimodal AI.

CERTIFICATIONS & TRAINING

Artificial Intelligence & Machine Learning

- Intro to Deep Learning & Machine Learning | Kaggle (May 2026)
- Artificial Intelligence Foundations: Machine Learning | LinkedIn Learning (Jun 2026)
- Machine Learning Foundations: Statistics | LinkedIn Learning (Jun 2026)
- Introduction to Prompt Engineering for Generative AI | LinkedIn Learning (Jun 2026)

Data Engineering & Analytics

- Advanced SQL & Intro to SQL | Kaggle (Jun 2026)
- Data Manipulation with Pandas | Kaggle (Jun 2026)

Software Development & Tools

- Git Essential Training | LinkedIn Learning (Jun 2026)
- Intro to Programming (Python) | Kaggle (May 2026)

Other

- Industry Preparation Training Programme 2025 | Faculty of Computing, NSBM Green University (July – Aug 2025)

SOFT SKILLS

- Complex Problem Solving & Analytical Thinking
- User Interface Design & User-Centric Problem Solving
- Academic Research & Technical Documentation
- Agile Development & Effective Time Management

LANGUAGES

- English (Professional Working Proficiency)
- Sinhala (Native)

REFERENCES

Dr. Rasika Ranaweera
Dean, Faculty of Computing,
NSBM Green University.
Email: ranaweera.r@nsbm.ac.lk

Ms. Sanuli Weerasinghe
Lecturer, Faculty of Computing
NSBM Green University
Email: sanuli.w@nsbm.ac.lk